

Information Sensors

Practical work (lab 6) report

“Distance and proximity sensors,
Environmental Data Sensors”

Student's first and last name in English: Zhu Chenhao

HDU ID: 22320630

ITMO ID: 375462

Hangzhou Dianzi University

2025

I. Scheme

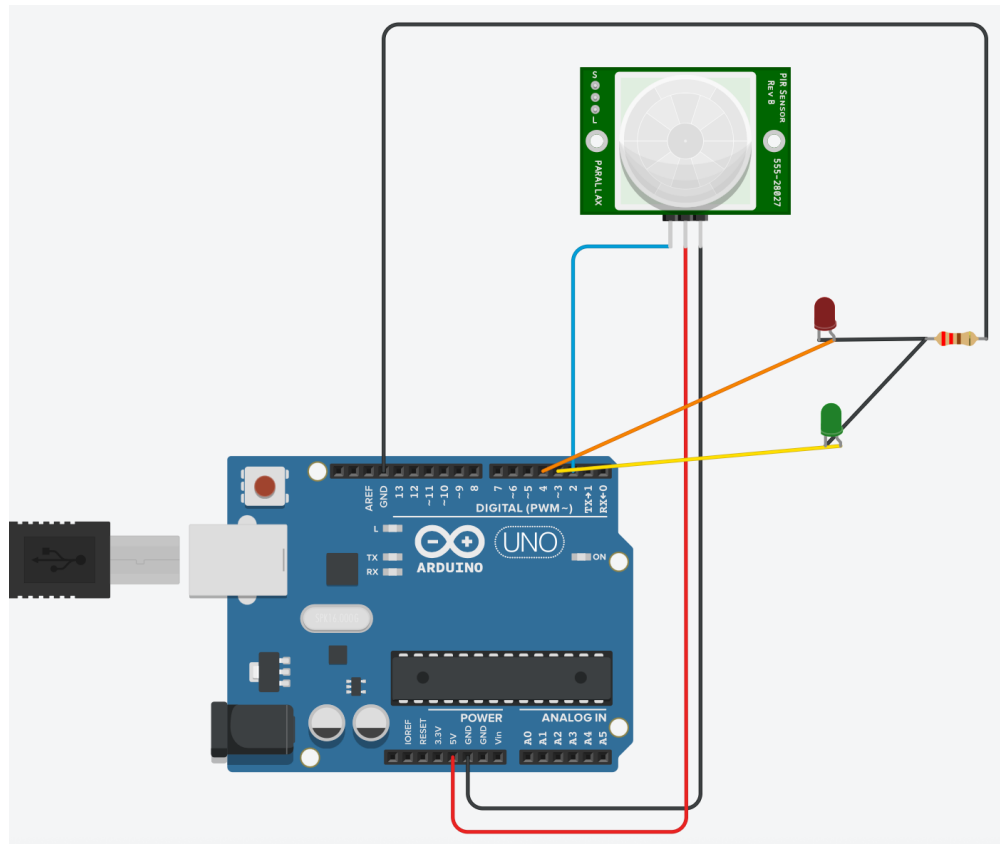


Figure 1. Schematic diagram of a PIR sensor with LED

II. Scheme parameters:

III. $T_{pulse} = t2(\text{sensor turn - off time}) - t1(\text{sensor turn - on time})$

$$T_{pulse} = 5.608 - 5 = 0.608 \text{ s}$$

Scheme type: retrigger

IV. Link to the TinkerCAD project :

<https://www.tinkercad.com/things/5LgI6zOJVr3-lab6zhu-chenhao22320630part1?sharecode=7oPAaciWUM17hjsxk6kOZLPLs312357qonxa6C00k70>

V.

VI. Scheme

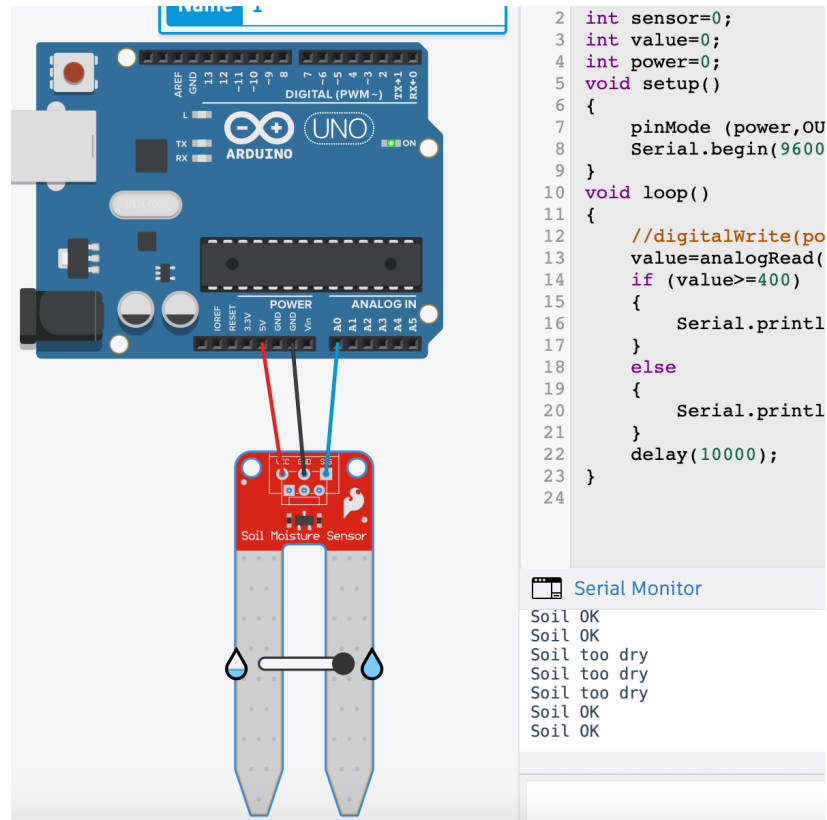


Figure 2. Schematic diagram of the soil moisture sensor

VII. Sensor characteristics

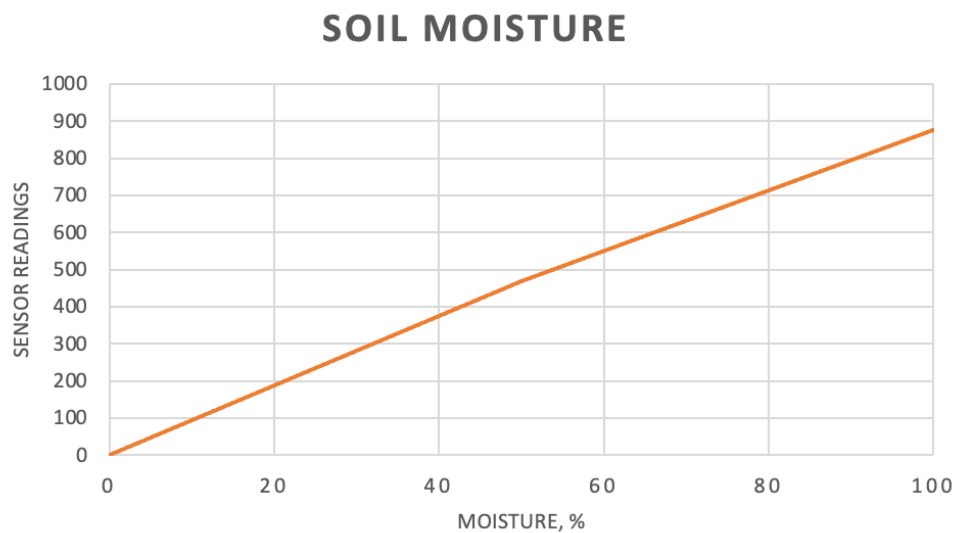


Figure 3. Soil moisture sensor characteristics graph

VIII. Sensor output:

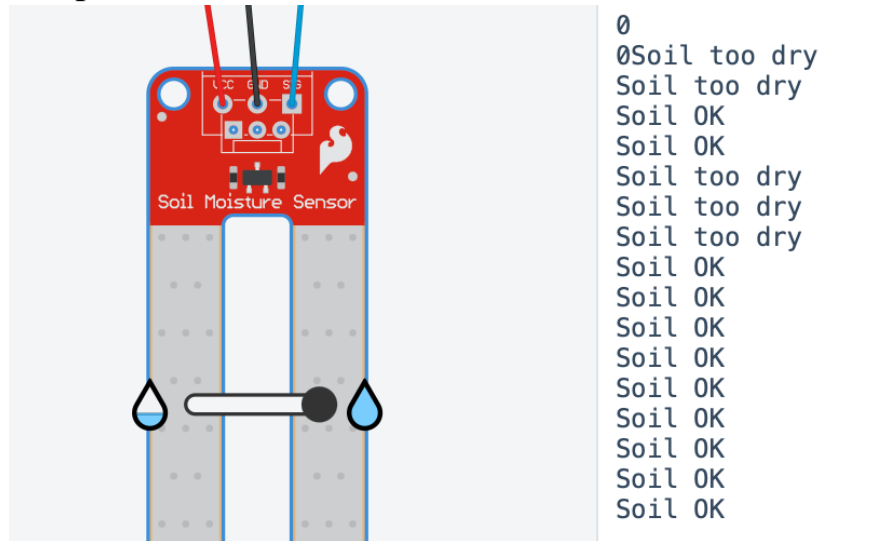


Figure 4. Soil moisture sensor output data

IX. Link to TinkercAD project : https://www.tinkercad.com/things/7mQmTZGB337-lab6zhu-chenhao22320630part3?sharecode=P55RQMpyLHe7WhsU1ulwhaeQpiZl_wPjg83Guig7-V8